Exp.no : 18 **MATPLOTLIB LIBRARY**

AIM:

To experiment to understand Matplotlib library use cases in Data Science through visualization

PYTHON CODE:

```
import matplotlib.pyplot as plt

# Sample dataset
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May',
          'Jun', 'Jul', 'Aug', 'Sep', 'Oct']

sales = [100, 120, 150, 130, 180, 200, 170, 220, 240, 260]
products = ['Laptop', 'Mobile', 'Tablet', 'Watch']
quantity = [30, 45, 20, 25]

# Create multiple visualizations
plt.figure(figsize=(12, 8))

# 1. Line Plot
plt.subplot(2, 2, 1)
plt.plot(months, sales, marker='o')
plt.title('Monthly Sales')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.xticks(rotation=45)

# 2. Bar Chart
plt.subplot(2, 2, 2)
plt.bar(products, quantity)
plt.title('Product Sales')
plt.xlabel('Product')
plt.ylabel('Quantity')

# 3. Pie Chart
plt.subplot(2, 2, 3)
plt.pie(quantity, labels=products, autopct='%1.1f%%')
plt.title('Sales Percentage')

# 4. Histogram
plt.subplot(2, 2, 4)
```

```
plt.hist(sales, bins=5, edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales')
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()
```

OUTPUT:



RESULT:

The code is executed successfully.